

OptiFlow Production Architecture & On-Premise Strategy

This document outlines the recommended production-grade technology stack for OptiFlow, specifically designed for seamless switching between Cloud and On-Premise deployments.

Recommended Full-Stack Architecture

Tier	Technology	Purpose
Frontend UI	React (Vite) + TailwindCSS	High-performance dashboard, fast rendering for large tables, responsive UI.
Web Server / Proxy	NGINX	Serves static frontend build, routes API traffic, handles HTTPS/SSL certificates.
Backend Framework	Python (FastAPI)	Async Python API layer; high performance, automatic OpenAPI documentation, strict Pydantic validation.
Database	PostgreSQL	Primary store for SOH, historical sales, store planograms, dispatch logs, and SKU catalog.
ORM & Migrations	SQLAlchemy + Alembic	Python object mapping and database schema version control.
Async Task Queue	Celery + Redis	Background queue for processing heavy allocation algorithms without freezing the UI.
Packaging & Deploy	Docker + Docker Compose	Wraps the entire stack into isolated containers for 1-click cloud or on-prem deployment.

Cloud vs. On-Premise Deployment Strategy

Because the entire application is containerized with Docker, the deployment architecture is identical in both environments.

1. Cloud Deployment (AWS / GCP / Azure)

- **Hosting:** The Docker containers run on a virtual cloud server (e.g., AWS EC2 or GCP Compute Engine).
- **Database:** Managed database service (e.g., AWS RDS PostgreSQL).
- **Access:** Accessible over the public web via secure HTTPS (<https://optiflow.yourdomain.com>).

2. On-Premise Deployment (Local Warehouse / Data Center)

- **Hosting:** The exact same Docker containers run on the client's internal physical server or local Virtual Machine (VMware / Proxmox).
 - **Database:** PostgreSQL runs inside a Docker container with local volume storage or connects to the client's internal PostgreSQL cluster.
 - **Access:** Accessible strictly over the company's internal network/Intranet (<http://192.168.1.50> or <http://optiflow.local>).
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Key Architectural Benefits

1. **Zero Licensing Costs:** The entire stack utilizes 100% open-source technologies with no recurring vendor software fees.
2. **Non-Blocking User Experience:** Heavy allocation algorithms run asynchronously in background Celery workers, keeping the manager dashboard fast and responsive.
3. **Data Integrity & Scale:** PostgreSQL provides ACID compliance for inventory movements and scales to handle millions of historical sales records.
4. **Instant On-Premise Portability:** Moving from Cloud to On-Premise requires zero code modifications—only environment configurations.